



Research Paper

Dual-temperature control for two storage chambers in a cold-chain logistics vehicle: An experimental study

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ABSTRACT

Cold-chain logistics (CCL) ensures stable transportation for products with different and strict storage temperature requirements. As temperature monitoring and control are crucial for such products, studies in CCL-related fields are expected to focus on developing refrigeration systems with adjustable temperatures or multi-temperature zones and analyzing the temperature characteristics of refrigerated chambers. Therefore, this study developed a low-order mathematical model with dual-temperature control for two refrigeration chambers and implemented it in a CCL vehicle to minimize temperature fluctuations during the storage period and accelerate temperature recovery to the required level after the chamber door is opened and closed. The linear matrix inequality algorithm and the H_2/H_∞ robust control method were employed to determine the optimal control gain. We performed relevant experimental tests and analyzed the system characteristics to validate the proposed control strategy. The temperature responses of the storage chambers from the present experiments were also compared with the results of the commercial CCL vehicle. The results demonstrated that the average temperatures of both chambers can be maintained at the specified levels after the system stability is achieved, with a maximum variation of $<0.5^\circ\text{C}$ to the target temperature. Moreover, the maximum oscillation of the chamber temperature was $0.4\text{--}2.2^\circ\text{C}$, successfully overcoming the shortcomings of excessive temperature oscillations ($4.5\text{--}8.5^\circ\text{C}$) in the commercial CCL vehicle used for comparison. The proposed system allows chamber temperatures to quickly recover to the required level and stabilize after door-opening operations.

1. Introduction

Cold-chain logistics (CCL) provides stable transportation with different temperature requirements for agricultural products, frozen food, pharmaceutical products, and chemicals. CCL management places significant emphasis on temperature and humidity monitoring between processes to maintain product quality. Many studies have collectively focused on optimizing CCL operations to improve efficiency, reduce costs, and minimize environmental impacts [1–3]. Some researchers have investigated the impact of door opening and airflow dynamics on temperature stability to optimize thermal management in refrigerated chambers [4–6]. For example, Oury et al. [4] reported that the process of opening the door was one of the main reasons for the temperature increase inside the chamber, indicating that low-temperature items may be affected by the ambient temperature, causing quality loss. De Micheaux et al. [5] found that with a narrower curtain opening, the temperature inside the chamber was less affected by the intrusion of

external air. In addition, the heating rate inside the chamber and the flow rate of the entering air decreased. Taher et al. [6] numerically studied the temperature characteristics inside a refrigerated chamber under different operating conditions. The simulation results demonstrated that the door-opening width significantly affected the internal temperature of the chamber.

Luchini et al. [7] applied model predictive control (MPC) to optimize the temperature control and energy consumption of a food transportation truck with a refrigerated chamber. They investigated the heating effect of high-temperature ambient air in the chamber when the door was opened. The simulation results indicated that compared with the traditional proportional-integral (PI) control method, the proposed MPC method performs better and can maintain the temperature inside the chamber at its allowable value. In addition, more efficient control can be achieved, and energy consumption can be effectively reduced by monitoring and controlling the temperature of goods instead of air. The aforementioned literature indicates that opening the door of the

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refrigerated chamber is the main reason for the temperature rise inside the chamber, which may lead to temperature fluctuation or overshoot and quality loss of the stored items. The above studies focus on optimizing temperature control and energy efficiency in refrigerated chambers, particularly during door-opening events. However, most studies overlook door-opening effects in control strategies, underscoring a gap in system design and practical application.

Moreover, some studies have emphasized dynamic modeling of the control system to improve CCL efficiency. For example, Artuso et al. [8] developed and experimentally verified a dynamic model of a CCL vehicle. Their results revealed that the maximum temperature difference in the chamber between the simulations and experiments was only 0.3 °C, demonstrating the high reliability of their dynamic model. Fallmann et al. [9] established a low-order dynamic model of small-scale CCL vehicles. This model has a centralized parameter system that considers temperature changes and power consumption. The system also considers the effects of opening doors on the environment and air inside the refrigerated chamber. The similarity between the simulated temperature results and the experiment was 86.2 %, indicating the applicability of their dynamic model. Shafiei and Alleyne [10] investigated the application of a thermal energy storage device combined with a vapor-compression refrigeration cycle in the CCL field. The CCL system was optimized by controlling the traffic conditions and load prediction. Through operating-load prediction, the size of the device can be reduced by 40 %, which can reduce the overall weight of the device and the required space. In addition, MPC can determine the best way to control the compressor, improving the system performance by 22 %.

Zhao et al. [11] conducted a literature survey on the application of refrigeration technologies in CCL and pointed out that future studies in CCL-related fields might focus on developing refrigeration systems with adjustable temperature or multi-temperature zones and analyzing the temperature characteristics of refrigerated chambers. Behdani et al. [12] also mentioned that an excellent temperature control technology can ensure product quality and reduce waste, reducing the operating costs of the entire CCL process. Temperature monitoring and control are important for products with strict storage temperatures. With the advancement of logistics service technology and the diversification of home delivery goods, different types of goods (with different temperature requirements) are typically delivered by the same vehicle. Kuo and Chen [13] addressed this concern by developing a multi-temperature joint distribution CCL system that can transport goods with different storage temperature requirements. This system improves the utilization of transportation vehicles and ensures product safety and quality. The above-reviewed studies explore refrigeration technologies and temperature control strategies for CCL. Collectively, these studies address the interplay between efficiency, sustainability, and technological advancement in CCL systems.

As indicated by our literature survey, most existing CCL-related research typically involved numerical simulations with relatively few experimental studies. In addition, CCL systems with multi-temperature requirements have possible applications; however, relevant research reports are limited. This gap underscores the need for more practical research and application-focused investigations. We addressed these limitations by developing a low-order mathematical model with dual-temperature control for two chambers and implemented it in a CCL vehicle to validate the proposed control strategy. Subsequently, we performed relevant experimental tests and analyzed the system characteristics. The temperature responses of the storage chambers from the present experiments were also compared with the results of the commercial CCL vehicle. This study aimed to reduce the temperature fluctuation during the storage period and accelerate the return of the temperature of the refrigerated and frozen chambers to the required level after door opening and closing.

2. Experimental details, system identification and control scheme

2.1. Experimental details

Fig. 1 shows a schematic and photograph of the experimental setup of the proposed refrigeration system with two chambers, including its main components: one compressor, one condenser, two throttle valves, and two evaporators. The working fluid (refrigerant) of this system is R404A. In addition, an oil separator is installed at the compressor outlet to separate the lubricating oil from the compressed vapor and guide it back to the compressor. Moreover, a filter and refrigerant sight glass are installed at the condenser outlet to ensure that the working fluid is free of impurities and completely condensed into a liquid state. Each chamber comprises a set of a throttle valve and an evaporator. A storage tank is installed between the evaporators and compressor to mix the working fluids outputted by the evaporator outlets at different pressures and temperatures. The system is also equipped with two control valves, one at the exit of the refrigerant sight glass and the other at the storage tank entrance, to control the flow of the working fluid. Finally, to avoid the dangers of unforeseen mishaps, the system was equipped with three safety valves connected with pipelines (dashed lines in Fig. 1) to instantly control the flow direction of the working fluid.

The total volume of the chambers intended for application in a CCL vehicle is 2.14 m³. According to our estimation, a total cooling capacity of approximately 1 kW is required to achieve and maintain a chamber temperature of − 25 °C under an environmental temperature of 35 °C. The condenser used in this study, with dimensions of 753 mm (length) × 305 mm (width) × 32 mm (height), was a microchannel heat sink made of aluminum alloy. In addition, the evaporators, with dimensions of 180 mm (length) × 400 mm (width) × 228 mm (height), were finned tube heat exchangers composed of copper (tube) and aluminum alloy (fin). The air in the external environment and inside the chambers was driven through the condenser and evaporators using fans. A scroll-type compressor (SIERRA03-0982Z1; Masterflux by Tecumseh) with a rotational speed of 0–6000 rpm and a displacement volume of 16 cm³ was used; however, it was operated below 5000 rpm during the experiments for safety reasons. The opening range of the throttle valves (UKV14D; Saginomiya Seisakusho) was 0–500 with a nominal capacity of 4.1 kW (for R404A, the working fluid in the present system). Several T-type thermocouples with uncertainty of 0.2 °C and pressure sensors with uncertainty of 3 % (XSK-AC30B-1036040; Saginomiya Seisakusho) were installed at the required locations, such as the inlet and outlet of the main components, to measure the temperature and pressure, respectively, as depicted in Fig. 1(a). A compact condenser box was installed on the outer of the chamber top. In this box, most of the components used in the present system were arranged inside, as shown in Fig. 1(b). In addition, the control and measurement system were mainly performed by a programmable logic controller system (FBs-24MC and FBs-6AD; FATEK Automation).

2.2. System identification and control scheme

System identification is used to infer the discrete governing equations of dynamic systems. This process is used to determine the relationship between the output responses and input commands of a controlled system, typically called the transfer function. Figs. 2 and 3 show the flowcharts of the system identification [14] and control scheme, respectively. In this study, we employ discrete-time state space representations in practical applications. Figs. 2 and 3 show the H matrix (for the outer loop), which represents the relationship between chamber temperature, fluid saturation temperature, and fluid superheat at the evaporator outlet. The G matrix (for the inner loop) represents the relationship between the fluid saturation temperature, fluid superheat at the evaporator outlet, compressor rotational speed, and opening of the throttle valve. With disturbance and anti-windup conditions, the state

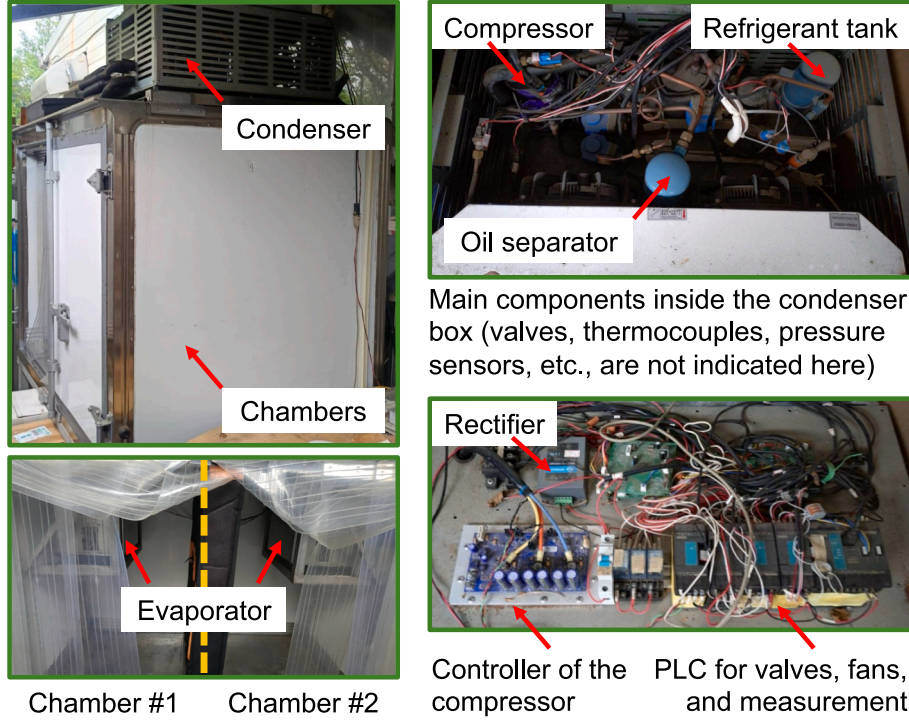
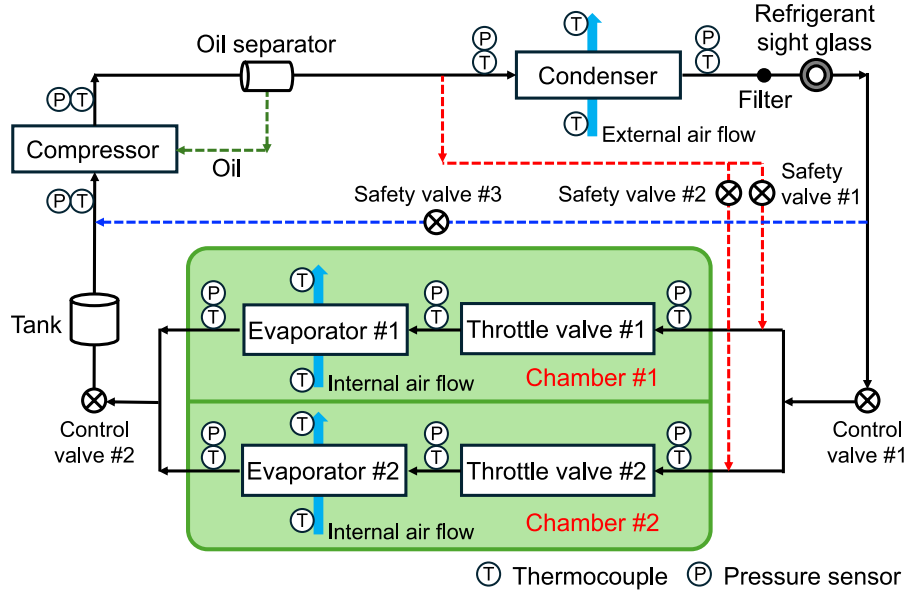


Fig. 1. Schematic and photograph of the experimental setup.

space equations for the outer loop can be formulated as follows:

$$X_o(k+1) = A_o X_o(k) + B_{o2} u_{os}(k) + B_{o\eta} u_{o\eta}(k) + B_{o1} w_o(k) \quad (1)$$

$$z_o(k) = C_{o1} X_o(k) + C_{o2} u_{os}(k) \quad (2)$$

where the subscripts o , s , and η represent the outer loop parameter, system, and anti-windup, respectively; X_o represents the state variables, such as the fluid saturation temperature and fluid superheat at the evaporator outlet; u_o represents the control inputs, including feedback

control gain and anti-windup; w_o is the outer loop disturbance; z_o represents the reference outputs such as the chamber temperature; A and B are constant matrixes; and $k = 1, 2, 3 \dots$. The discrete-time dynamic equation for the inner loop can be expressed as the following state space form:

$$X_i(k+1) = A_i X_i(k) + B_{i2} u_{is}(k) + B_{i\eta} u_{i\eta}(k) + B_{i1} w_i(k) \quad (3)$$

$$z_i(k) = C_{i1} X_i(k) + C_{i2} u_{is}(k) \quad (4)$$

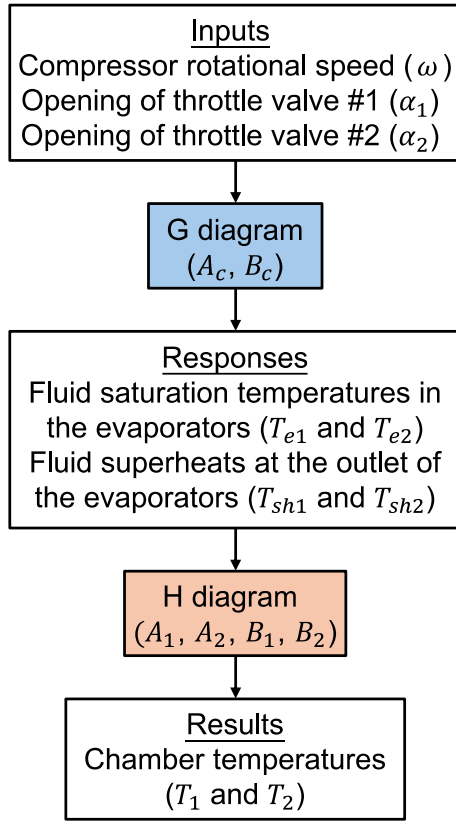


Fig. 2. Flowchart of the system identification process.

where the subscript i represents the inner loop parameter; X_i represents the state variables, such as the compressor rotational speed and opening of the throttle valve; u_i represents the control inputs; w_i is the inner loop disturbance; z_i is the reference outputs, such as the fluid saturation temperature and fluid superheat at the evaporator outlet.

The control system sends a command signal to the compressor and electrical throttle valves (i.e., control variables) and obtains the corresponding responses regarding the fluid saturation temperature, fluid superheat at the evaporator outlets, and chamber temperatures. The optimal gain value was calculated, and a reduced-order algorithm was developed by employing a linear matrix inequality (LMI) algorithm [15] in the control panel of the CCL system to avoid differential items and determine the optimal PI control gain. After obtaining the governing equations, they must be discretized to align with the requirements for digital control. In this study, the H_2/H_∞ robust control method [16] was used to obtain the control gain algorithm. After applying the LMI, all discrete governing equations become first-order linear inequalities. If the optimal control gain can be obtained, the Lyapunov system will be asymptotically stable, making it easy to determine the stability of the system. In the practical application of the proposed system, the first-order control gain must only capture the sensor signals and then apply simple algebraic equations to determine the appropriate compressor rotational speed and throttle-valve openings to control the chamber temperatures. In addition, a feedback control with PI gains was implemented on the circuit board. For better understanding and further application by the reader, the detailed equations and descriptions for the H_∞ , H_2 , and LMI expressions can be found in our previous study [15].

As shown in Fig. 4, we used three specific commands based on the system identification results. These include the compressor rotational speed (ω), opening of the first throttle valve (α_1), and opening of the second throttle valve (α_2). The total test duration of 240 min was divided

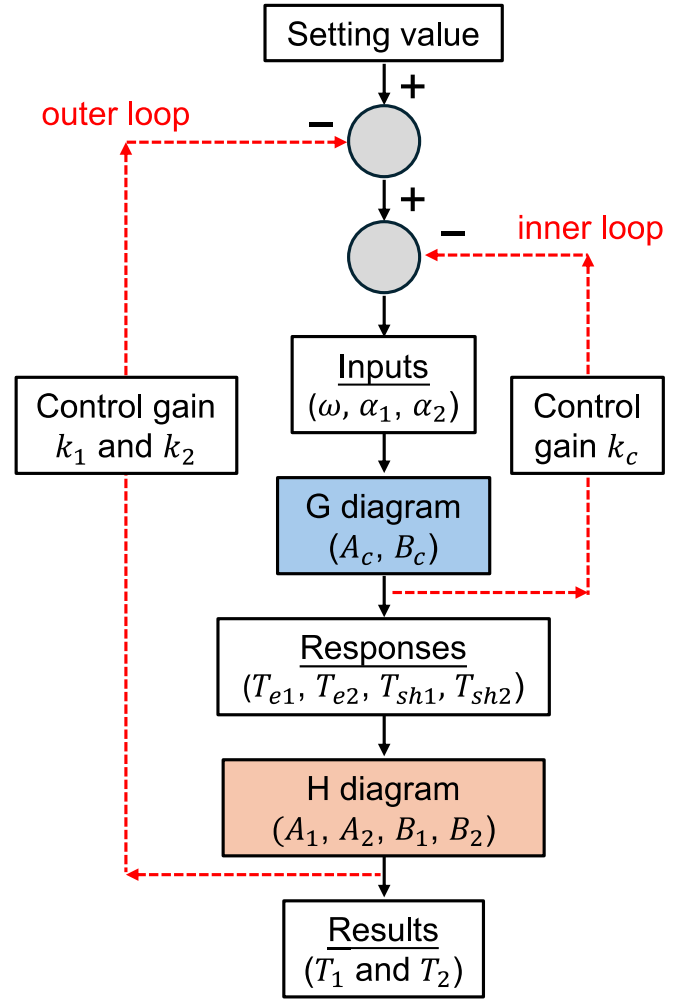


Fig. 3. Flowchart of the system feedback control.

into three 80-min periods. During the first period (0–80 min), α_1 was changed; however, α_2 and ω were maintained at 300 and 2430 rpm, respectively. In the subsequent period (80–160 min), only α_2 was changed. Finally, only ω was changed in the last period (160–240 min). By contrast, the two throttle-valve openings were controlled at 300.

Fig. 5 shows the fluid saturation temperature in the first and second evaporators (T_{e1} and T_{e2} , respectively) and the fluid superheats at the outlets of the first and second evaporators (T_{sh1} and T_{sh2} , respectively). Subsequently, A_c and B_c in Equation (5) were determined based on the relationship between the compressor rotational speed, throttle-valve openings, and two sets of fluid saturation temperatures and superheat at the evaporators. Finally, we obtained the feedback-loop control gain k_c , as shown in Equation (9). The corresponding governing equations and results can be summarized as follows:

$$\dot{x} = A_c x + B_c u \quad (5)$$

$$u = -k_c x \quad (6)$$

$$x = \begin{bmatrix} T_{e1} & \dot{T}_{e1} & T_{e2} & \dot{T}_{e2} & T_{sh1} & \dot{T}_{sh1} & T_{sh2} & \dot{T}_{sh2} \end{bmatrix}^T \quad (7)$$

$$u = [\omega \quad \alpha_1 \quad \alpha_2]^T \quad (8)$$

$$k_c = \begin{bmatrix} -0.2651 & -197070 & -0.0136 & 1.5532 & -0.1549 & 94.2565 & -0.2279 & -4.4803 \\ 0.0014 & 4.3194 & 0 & 1.7532 & -0.0106 & -101.7725 & 0.0036 & 6.6709 \\ 0.0065 & 7.9128 & -0.0003 & -3.7693 & -0.0003 & 1.3706 & -0.0100 & -8.7220 \end{bmatrix} \quad (9)$$

The temperature profiles of both chambers (T_1 and T_2) are shown in Fig. 6. Using the same method and data presented in Figs. 5 and 6, A_j and B_j in Equation (10) can be obtained based on the relationship between the two sets of fluid saturation temperatures (T_{ej}), superheat (T_{shj}), and

chamber temperatures (T_j). Subsequently, the feedback-loop control gain k_j can be obtained, where $j = 1$ or 2 , indicating Chambers #1 or #2, respectively. The corresponding governing equations and results can be summarized as follows:

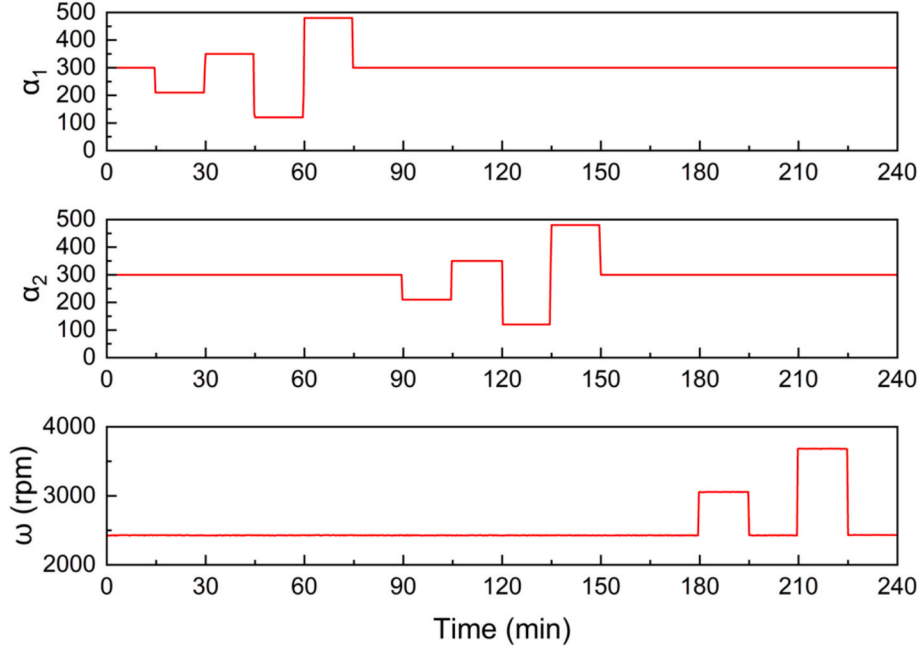


Fig. 4. Control variables of the system identification experiment: throttle-valve openings and compressor rotational speed.

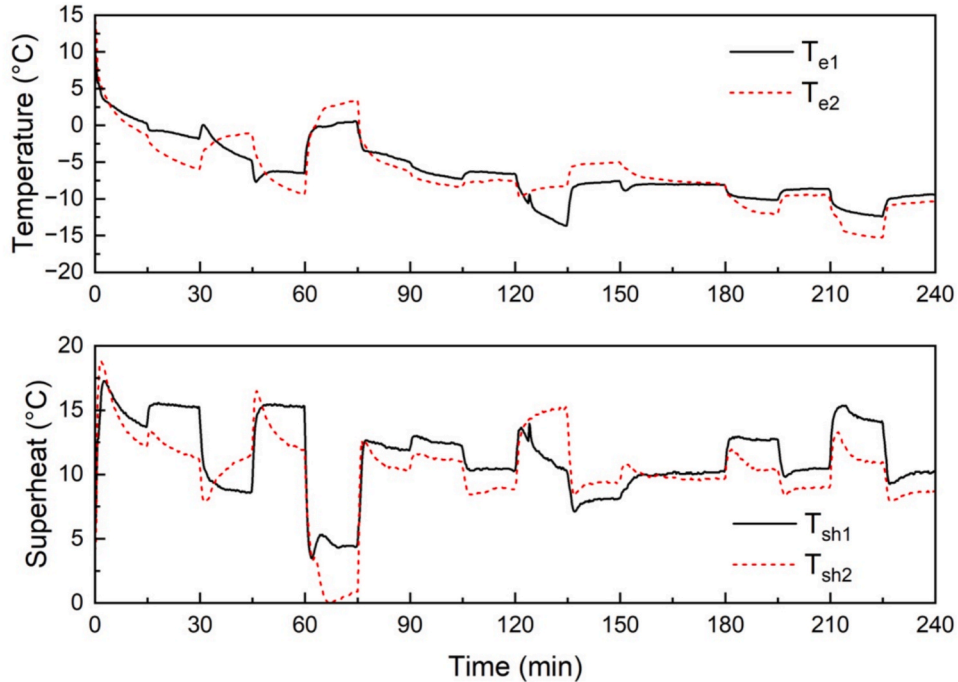


Fig. 5. Responses of the system identification experiment: fluid saturation temperatures and superheats at the evaporator outlet.

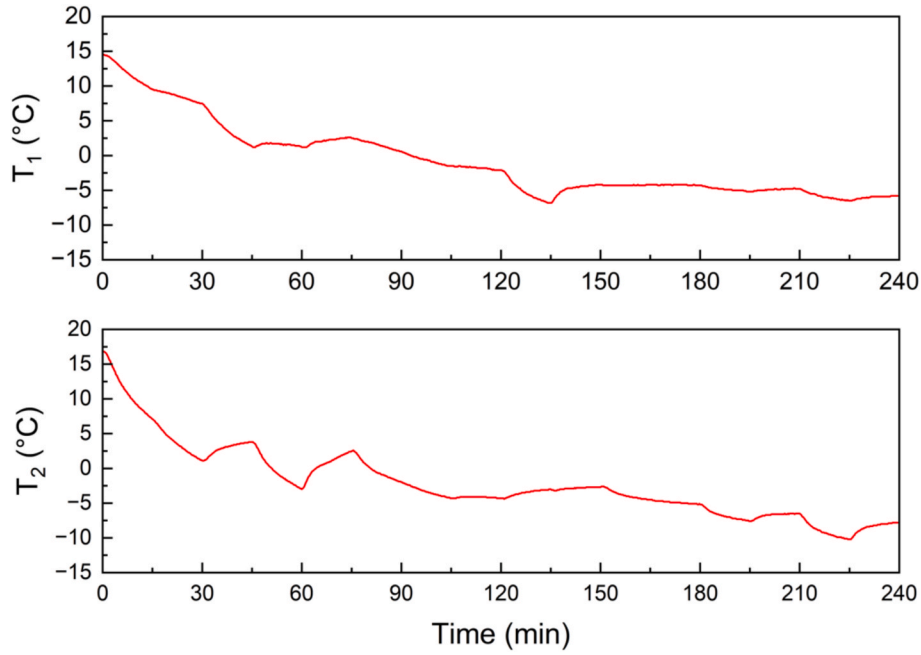


Fig. 6. Responses of the system identification experiment: temperature profiles of Chambers #1 and #2.

$$\dot{x}_j = A_j x_j + B_j u_j \quad (10)$$

$$u_j = -k_j x_j \quad (11)$$

$$x_j = \begin{bmatrix} T_j & \dot{T}_j \end{bmatrix}^T \quad (12)$$

$$u_j = \begin{bmatrix} T_{ej} & T_{shj} \end{bmatrix}^T \quad (13)$$

$$k_1 = \begin{bmatrix} 0.0173 & 4.6850 \\ 0.0430 & 11.6682 \end{bmatrix} \text{ and } k_2 = \begin{bmatrix} 0.0387 & 5.3092 \\ 0.1325 & 18724 \end{bmatrix} \quad (14)$$

3. Results and discussion

3.1. Commercial CCL vehicle

The performance of the proposed system was compared with that of a commercial CCL vehicle comprising two storage chambers (total chamber volume = 2.67 m³) with different temperature requirements similar to that of the proposed system. One chamber is referred to as refrigerated (Chamber R), and the other as frozen (Chamber F). The ambient temperature in the non-controlled lab environment during the experiments was 24.3–25.7 °C. The refrigeration system of this commercial CCL vehicle was only implemented in the frozen chamber (with the temperature set to −15 °C). The temperature of the refrigerated chamber (4 °C) was controlled using a fan to introduce air from the frozen chamber. Therefore, the temperature of the refrigerated chamber was highly influenced and limited by the frozen chamber condition. Owing to confidentiality restrictions, we could not determine the mathematical model and control strategy of the refrigeration system employed in this vehicle.

Fig. 7 presents the experimental results of the frozen chamber, including the frozen chamber temperature (T_F) and evaporator outlet temperature ($T_{F,out}$). Evidently, in the early stages (0–25 min), T_F and $T_{F,out}$ rapidly dropped from 26 °C to −10 and −15 °C, respectively. Subsequently, once $T_{F,out}$ reached the target value, the refrigeration system briefly reduced the compressor operating output and maintained it for approximately 15 min. However, because T_F did not reach the target value, the refrigeration system gradually increased the

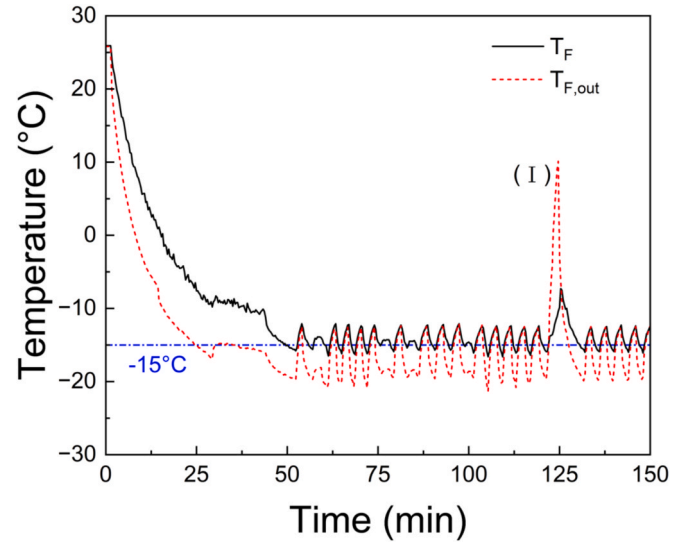


Fig. 7. Experimental results of the frozen chamber in the commercial CCL vehicle.

compressor operating output. After approximately 50 min of operation, T_F successfully reached the target of −15 °C; however, $T_{F,out}$ reached −20 °C. Thus, for a target temperature of −15 °C, the T_F and $T_{F,out}$ in this commercial CCL vehicle differed by 5 °C. Subsequently, during the next 60 min (i.e., a test interval of 60–120 min), T_F and $T_{F,out}$ both exhibited similar and stable oscillation characteristics in approximately 5-min periods, during which the average, maximum, and minimum T_F values were −14.3, −12.1, and −16.6 °C, respectively, indicating a maximum temperature oscillation of 4.5 °C. In addition, the average, maximum, and minimum $T_{F,out}$ were −17.0, −12.1, and −21.3 °C, respectively, exhibiting a maximum temperature oscillation of 9.2 °C. These results indicate that the average T_F varied by 0.7 °C from the target, demonstrating that this commercial CCL vehicle satisfied the required operating conditions. After 120 min of continuous testing, the system conducted automatic defrosting for approximately 3 min (i.e., during the test period of

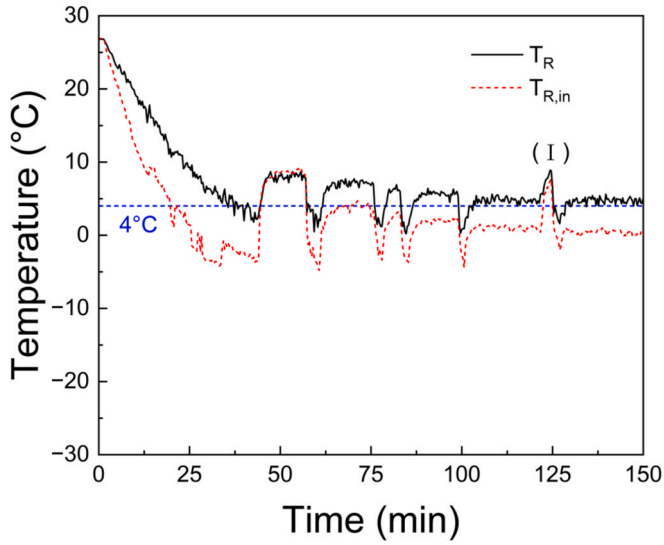


Fig. 8. Experimental results of the refrigerated chamber in the commercial CCL.

120–123 min). The experimental results showed that when defrosting began, T_F and $T_{F,out}$ rapidly increased to 10 and -7.5°C , respectively, in a short period. After defrosting, both temperatures dropped to their original stable and oscillating temperature range within 10 min (marked as Interval I in the figure).

Fig. 8 shows the experimental results of the refrigerated chamber, including the temperatures of the refrigerated chamber (T_R) and the air introduced by the fan ($T_{R,in}$). Fig. 8 indicates that T_R and $T_{R,in}$ both dropped significantly at the beginning of the test and reached the target temperature (4°C) after approximately 20 and 40 min, respectively. However, from 20 to 40 min, T_R approached the target value slowly, and the fan continuously introduced air from the frozen chamber to cool it. Therefore, $T_{R,in}$ kept decreasing (owing to the decrease in T_F at this time, as shown in Fig. 7) and approached its lowest value of -5°C . During the 50–120 min test interval, T_R and $T_{R,in}$ exhibited similar trends; however,

no periodic characteristics were evident. During this period, the average, maximum, and minimum values of T_R were 5.3 , 8.7 , and 0.2°C , respectively, indicating a maximum oscillation of 8.5°C . This was significantly higher than that in the frozen chamber, as presented previously. In addition, the average, maximum, and minimum values of $T_{R,in}$ were 2.2 , 9.2 , and -4.8°C , respectively, indicating a maximum oscillation of 14°C . Furthermore, during the test interval of 110–150 min, excluding the influence of the automatic defrosting process of the frozen chamber (i.e., Interval I), T_R and $T_{R,in}$ were maintained at 4 and 0°C , respectively. From the above results, we concluded that introducing cold air from the frozen chamber through a fan to control the temperature of the refrigerated chamber has its limitations, especially during the early stages of system operation. More importantly, even under relatively stable operating conditions, the temperatures of both chambers exhibited significant fluctuations, which would inevitably affect the quality of the products stored in them.

3.2. Proposed system

Fig. 9 presents the experimental results of the proposed system during the 0–180 min test interval, including the chamber temperatures (T_1 and T_2) and evaporator outlet temperatures ($T_{1,out}$ and $T_{2,out}$). The temperatures of Chambers #1 (refrigerated chamber) and #2 (frozen chamber) were set to 8 and -2°C , respectively, and the ambient temperature was about 19°C . In addition, the doors of both chambers were simultaneously opened for 1 min after 127 min. This door-opening operation simulated the impact of the external environment on the internal temperatures of the chambers when logistics and distribution personnel open doors to pick up or place items. The experimental results showed that T_1 and $T_{1,out}$ were approximately similar, with a difference of $<1^\circ\text{C}$, indicating that T_1 could be effectively controlled and maintained. Moreover, both temperatures reached the target values after approximately 15 min. Subsequently, both temperatures remained stable without any noticeable fluctuations. They increased slightly by 1°C during the 1-min door-opening operation and then quickly returned to the target value within 3 min; in addition, the impact of the temperature fluctuation lasted for <5 min. The 90–120 min interval results indicate that the average, maximum, and minimum values of T_1 were 7.5 (difference of only 0.5°C from the setting), 7.7 , and 7.3°C , respectively,

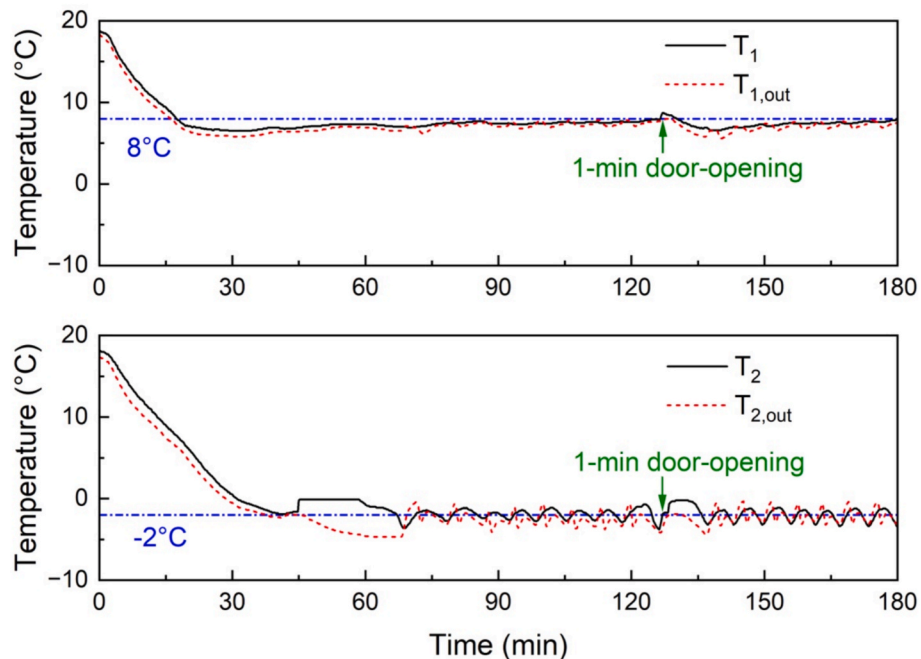


Fig. 9. Experimental results of the proposed system during the test interval of 0–180 min.

indicating a maximum oscillation of 0.4 °C. Furthermore, the average, maximum, and minimum $T_{1,out}$ values were 7.3, 7.9, and 6.8 °C, respectively, indicating a maximum oscillation of 1.1 °C. Thus, the oscillation in Chamber #1 was significantly lower than that in the commercial CCL vehicle, demonstrating the advantages of the proposed system configuration and control strategy.

Regarding Chamber #2, the results showed that T_2 reached the target value (−2 °C) after 45 min. Subsequently, unlike T_1 and $T_{1,out}$, T_2 and $T_{2,out}$ exhibited similar and stable oscillation characteristics, particularly from 70 to 180 min. In addition, the temperature oscillation exhibited a specific cycle (approximately 6 min), similar to the commercial CCL vehicle. Moreover, similar to Chamber #1, a short door-opening operation at 127 min caused T_2 to rise slightly by 2 °C; however, it quickly returned to the target value. The overall impact time on the system was approximately 10 min. Considering the stable oscillation range between 90 and 120 min as an example, the average, maximum, and minimum values of T_2 were −2.0 (exactly equal to the target value), −1.1, and −2.9 °C, respectively, indicating a maximum temperature oscillation of 1.8 °C. In addition, the average, maximum, and minimum values of $T_{2,out}$ were −2.5, −0.6, and −3.8 °C, respectively, indicating a maximum temperature oscillation of 3.2 °C. Similar to the results for Chamber #1, the difference between T_2 and $T_{2,out}$ was small. These temperature oscillations and differences were significantly lower than those in the commercial CCL vehicle, again validating the advantages of the proposed system.

To further understand the operating characteristics of the present system, Fig. 10 shows the relationship between the control parameters of the refrigeration system as a function of time, including the rotational speed of the compressor and the opening of the throttle valves in the chambers. In particular, the figure shows that during the initial period (0–15 min), the compressor was operated at a high rotational speed (5000 rpm) because the chamber temperatures were the same as the ambient temperature. In addition, both throttle valves were simultaneously opened to a high level (480) to satisfy the requirement for rapid cooling of both chambers. Subsequently, when the system was relatively stable during 70–180 min, the compressor rotational speed was controlled between 1500 and 3000 rpm, and the opening of the throttle valves in Chambers #1 and #2 was controlled from 100 to 400 and 300–450 rpm, respectively, thereby exhibiting a similar trend.

Fig. 11 shows the results for the test interval of 180–380 min. We explored the impact of different temperature settings by adjusting the

target temperature of Chamber #1 from 8 to −3 °C at 195 min and that of Chamber #2 from −2 to −15 °C. Subsequently, a 3-min door-opening operation was performed at 340 min. The results showed that Chamber #1 required approximately 60 min to reach the new target temperature, whereas Chamber #2 took considerably longer (approximately 120 min). These results indicate that when the system operated under steady conditions and the temperature settings of the chambers were suddenly changed, reaching the new target temperatures took a relatively long time. The dynamic response characteristics of this system can enable the development of important operating guidelines for users, i.e., before the CCL is implemented, the storage temperature requirements for transporting the products must be clearly understood, and the settings employed accordingly. In addition, the temperature settings should not be changed during transportation. The results also demonstrated that the differences between the temperatures of the two chambers and their corresponding evaporator outlet temperatures were small. The temperature of Chamber #1 was maintained at a stable value after 270 min, except for the temperature fluctuation caused by the door-opening operation at 340 min. Considering the data from 350 to 380 min (the last 30 min of the experiment) as an example, the average temperature of Chamber #1 was −3.3 °C, indicating a difference of 0.3 °C from the target value. Furthermore, the maximum difference between the high and low temperatures was 0.8 °C. Similar to the experimental results for 0–180 min, the temperature of Chamber #2 exhibited stable oscillation after reaching the new target value. In addition, after the door-opening operation at 340 min, the temperature of Chamber #2 increased from −15 to −10 °C, and the impact of temperature fluctuations persisted for approximately 15 min. The results also indicated that the longer the door is kept open, the greater the temperature rise in the chamber and the longer it takes for the system to return to its original stable state. Moreover, the results of the last 30 min demonstrated that the average temperature of Chamber #2 was −14.6 °C (indicating a small temperature deviation from the target value). The maximum difference between the high and low temperatures was 2.2 °C.

In conclusion, the experimental results of this study demonstrated that the average temperatures of both chambers can be maintained at the specified levels, with a maximum difference of <0.5 °C to the target temperature, for extended periods after the system stabilizes. Furthermore, the chamber temperature exhibited a maximum oscillation of 0.4–2.2 °C, demonstrating that it can overcome the shortcomings of excessive temperature oscillations prevalent in the commercial CCL

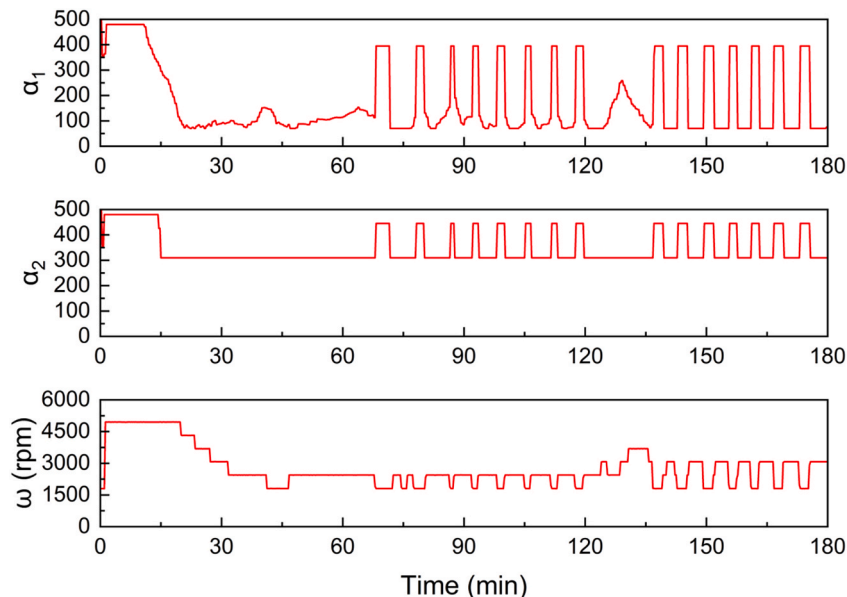


Fig. 10. Throttle-valve openings and compressor rotational speed in the proposed system during the test interval of 0–180 min.

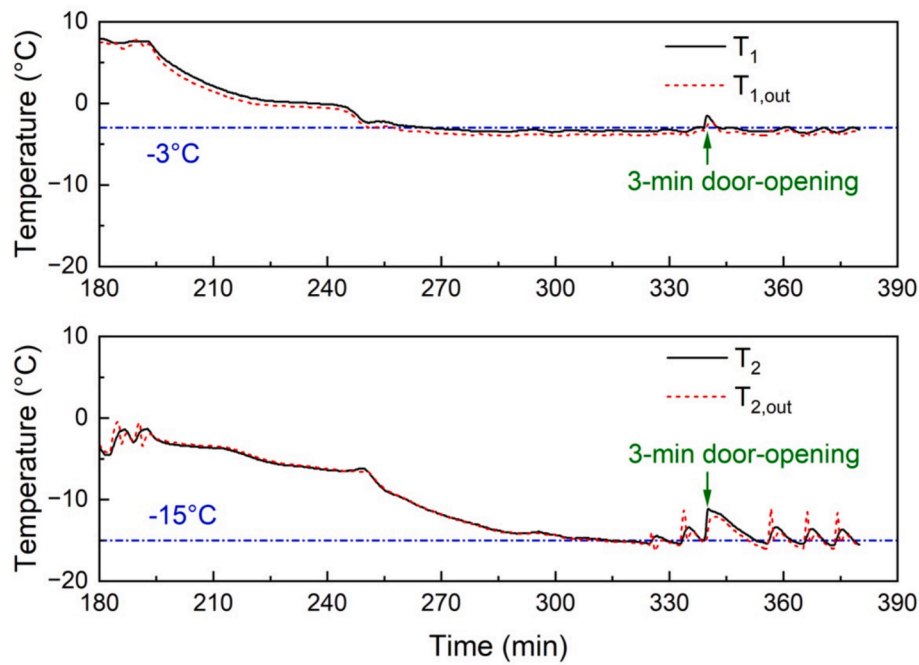


Fig. 11. Experimental results of the proposed system during the test interval of 180–390 min.

Table 1

Comparison between the commercial and proposed systems.

	Commercial system Chamber R	Chamber F	Proposed system Chamber #1	Chamber #2	Improvement of the proposed system
(1) Temperature control method for the chamber	Cold air from Chamber F	A refrigeration system	A refrigeration system	A refrigeration system	Dual-temperature control
(2) Target temperature of the chamber	4 °C	−15 °C	8 °C (or − 3 °C)	−2 °C (or − 15 °C)	Adjustable target value separately
(3) Average temperature of the chamber under the stable period	5.3 °C	−14.3 °C	7.5 °C (or − 3.3 °C)	−2 °C (or − 14.6 °C)	Much closer to the target value
(4) Difference of the average and target temperatures under the stable period	1.3 °C	0.7 °C	0.5 °C (or 0.3 °C)	0 °C (or 0.4 °C)	Difference was reduced by 0.3–0.8 °C
(5) Temperature fluctuation of the chamber	8.5 °C	4.5 °C	0.4 – 0.8 °C	1.8 – 2.2 °C	Fluctuation was reduced by 2.7–8.1 °C
(6) Recovery time to reach the target temperature after the door-opening operation	35 min	20 min	5 min	10–15 min	Required time was shortened by 5–30 min

vehicle used for the comparisons, which exhibited oscillations of 4.5–8.5 °C. Finally, the proposed system can return the chamber temperatures to the target settings rapidly and stabilize them after door-opening operations. Table 1 summarizes the comparison between the commercial and proposed systems. The improvement of the proposed system has also been provided in this table. For the commercial CCL system, only the temperature of the frozen chamber was controlled by the refrigeration system of the vehicle; the temperature of the refrigerated chamber was controlled by the cold air driven from the frozen chamber. Therefore, the temperature fluctuation of the refrigerated chamber was much higher than that of the frozen chamber. On the contrary, we have successfully developed dual-temperature control for two chambers with an adjustable target value separately. Compared with the commercial system, for the proposed system and during the stable period, the average chamber temperature was much closer to the target value, for instance, the difference to the target value was reduced by 0.3–0.8 °C; the temperature fluctuation of the chamber was also reduced by 2.7–8.1 °C; and the required time to return to the target temperature after the door-opening operation was shortened by 5–30 min.

4. Conclusions

We developed a low-order mathematical model with dual-temperature control for two chambers and implemented it in a CCL vehicle to minimize temperature fluctuations during the storage period and accelerate the temperature recovery in both chambers to the target levels after door opening and closing operations. We performed relevant experimental tests and analyzed the system characteristics to validate the proposed control strategy. The experimental results of the proposed system demonstrated that the chamber and evaporator outlet temperatures were almost identical, indicating that the temperatures of both chambers could be effectively controlled and maintained. Under relatively stable conditions of the initial experimental period, the maximum temperature oscillations of Chambers #1 and #2 were only 0.4 and 1.8 °C, respectively. During the subsequent experimental period, the target temperatures of both chambers were suddenly changed to the lower levels, which required relatively longer times to achieve the new target temperatures. The results of both experimental periods demonstrated that the average temperatures of both chambers could be maintained for extensive periods with a maximum difference of <0.5 °C to the target temperature after the system stabilizes. The maximum temperature fluctuations were only 0.4–2.2 °C, overcoming the issue of

excessive temperature fluctuations (4.5–8.5 °C) in the commercial CCL vehicle used for comparisons. In addition, the proposed system ensures the restoration of chamber temperatures to their target values and stabilization after a door-opening operation, thus advancing temperature monitoring and control in CCL.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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